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Family Replacement Donation as a Predictor of Hepatitis B Virus Infection Among Asymptomatic Blood Donors in Tanzania

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Abstract

Background: Blood transfusions save lives, but transfusing unsafe blood risks exposure to transfusion-transmissible infections such as Human Immunodeficiency Virus, Hepatitis B, Hepatitis C, Syphilis, malaria, and other infectious agents. In Tanzania, all blood donors are screened using a blood donor questionnaire, and only presumed healthy individuals are permitted to donate blood after pre-donation counselling. This study was conducted to investigate predictors of Hepatitis B infection among asymptomatic blood donors.

Methodology: We conducted a cross-sectional study at the National Blood Transfusion Eastern Zonal Center. Convenient sampling was employed to select study participants by reviewing records from the Blood Establishment Computer System (BECS) database of blood donors who donated blood at the fixed site from January 2018 to December 2019. Data analysis was conducted using SPSS version 23; descriptive statistics were summarized using frequencies and percentages. A p-value of <0.05 was considered significant.

Results: The overall prevalence of Hepatitis B infection was 5.4%. A higher prevalence was observed among family replacement donors (8.2%) than among voluntary donors (4.1%) (p<0.001). Family replacement donation was shown to be a significant indicator of Hepatitis B infection (aOR=2.1, 95%CI= 2.1(1.64-2.58), (P< 0.001).

Conclusion: Family replacement donation was a predictor of HBV infection. More efforts should be focused on encouraging voluntary blood donations, as they have been shown in this study, as well as previous studies to have a lower risk of Hepatitis B infection.

Keywords: Blood donation, Hepatitis B infection, Voluntary blood donor, Family replacement donors

Introduction

The Global Database on Blood Safety (GDBS) estimated that a total of 112.5 million blood donations were made in 180 countries worldwide [WHO., 2017]. A significant challenge in blood donation systems is blood wastage due to Transfusion-Transmissible Infections (TTIs), with Hepatitis B virus infection (HBV) being the leading cause of blood discard [WHO.2017, Patil et al. [2016]. Blood transfusion saves

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lives; however, unsafe blood transfusions can put lives at risk due to TTIs such as Human Immunodeficiency Virus (HIV), HBV, Hepatitis C (HCV), syphilis, malaria, and other infectious agents. In Tanzania, all blood donors are screened using a donor questionnaire prior to donation, and only presumed healthy individuals are permitted to donate blood [NBTS.2015].

Hepatitis B virus infection is a serious global health issue. In Western Europe and North America, the disease is relatively rare and is primarily acquired in middle age. In contrast, in Asia and most African countries, chronic HBV infection is more prevalent and is typically acquired prenatally or during childhood. The seroprevalence of HBV among the general population and blood donors varies between countries. It is reported to be 0.1-0.5% in the general population in the United States and Western Europe, whereas a prevalence rate of 5 to 20% has been reported in the Far East and certain tropical countries [Soomro et al.2011].

HBV infection is the leading cause of blood discarding. A study in Nigeria reported an HBV prevalence of 10% among blood donors, with males being more affected than females [Omosigho et al. 2022]. In Tanzania, Kilonzo et al. reported that HBV detection was at 6.2% and accounted for 42% of all TTIs reported by Tanzania's Blood Service Fact Sheet 2017 [Kilonzo et al.2018]. Family Replacement Donors (FRD) had a higher risk of infection at 21.5% compared to voluntary, unpaid donors, who had a risk of 12.7% [Kilonzo et al.2018].

According to the NBTS annual report (unpublished), HBV infection continues to be the most frequent TTI and the leading cause of blood discard [NBTS. 2020]. Previous reports have shown that HBV is the leading cause of blood discard, contributing to nearly half of all discards due to TTIs, with Family Replacement Blood Donors (FRBDs) being the most infected [MoH.2017]. The natural history of HBV infection among adults could contribute to these high rates of HBV infection among blood donors, as a person can transmit the infection during its subclinical phase. Due to its chronic sequelae, such as liver cirrhosis and hepatic cancer, there is a need to improve HBV detection rates.

Blood safety remains a significant concern in transfusion medicine in Tanzania, and several measures have been implemented to enhance transfusion safety. The NBTS has updated its screening donor questionnaire to improve the detection rate of donors at risk of TTIs. The blood donor questionnaire, utilized by blood donor counselors, has been the most revised document in blood safety, aimed at improving the identification of high-risk donors before donation. Currently, version 8 of the blood donor questionnaire includes questions specifically assessing HBV infection risk factors, which were absent in previous versions [NBTS.2019].

Despite these efforts, HBV prevalence has remained similar between blood donors (6.2%) and the general population (6%). It is the leading cause of blood discard compared to other TTIs [Kilonzo et al.2018]. The main question remains: How can high-risk blood donors with HBV infection be detected using a blood donor questionnaire before blood donation? Therefore, this study aimed to assist the NBTS in understanding the inherent risk factors for HBV infection among relatively healthy blood donors.

Methodology

Study design and setting

A descriptive retrospective cross-sectional study was conducted at the Eastern Zone National Blood Transfusion Center in the Ilala municipality of Dar es Salaam. The zone collects blood donations from Monday to Sunday and has two mobile blood collection teams, along with a fixed-site donation unit that accommodates walk-in blood donors.

Study population and study duration

We reviewed blood donor records from the Blood Donors Establishment Computer System (BECS) for all donors who donated blood at the fixed site from January 2018 to December 2019. The eastern zone was purposely selected because it represents a large proportion of blood donations per year compared to other zonal centers. It also has a higher discard rate due to HBV infection compared to other zonal centers [NBTS. 2020 unpubl]. Fixed-site donors were chosen because they are more likely to be notified of their previous blood donation TTI markers compared to mobile donors, as counsellors cannot access all previous donor results when outside the center.

Sample size and sampling method

A convenient sampling method was used to select study participants by reviewing all records from the database, specifically examining the records of clients who donated blood. We included blood donors with complete records and excluded those with incomplete information.

Data collection and analysis

Data were extracted from the BECS into an Excel sheet before being exported to SPSS. The extracted variables included Age, Sex, Residence, blood donation frequency, and type of blood donor. Data analysis was conducted using SPSS version 23. Descriptive statistics were summarized using frequencies and percentages. Odds ratios and 95% confidence intervals (CI) were calculated as appropriate, with a p-value of less than 0.05 considered statistically significant.

Ethical consideration

We received ethical approval from the Muhimbili University of Health and Allied Sciences (MUHAS) Institutional Review Board through the Department of Hematology and Blood Transfusion Research Committee (Ref.No.DA.25/111/01). Permission to conduct the study was sought from the National Blood Transfusion Services Authority. We used only electronically generated donation numbers without blood donor names to ensure anonymity and confidentiality.

Results

During the study, 7179 eligible blood donors donated blood at the NBTS eastern zonal center fixed site. We excluded 112 blood donors due to incomplete donor records (two missing pieces of information on donor age and 110 on blood donor type). The final analysis included 7067 donors (Figure 1).

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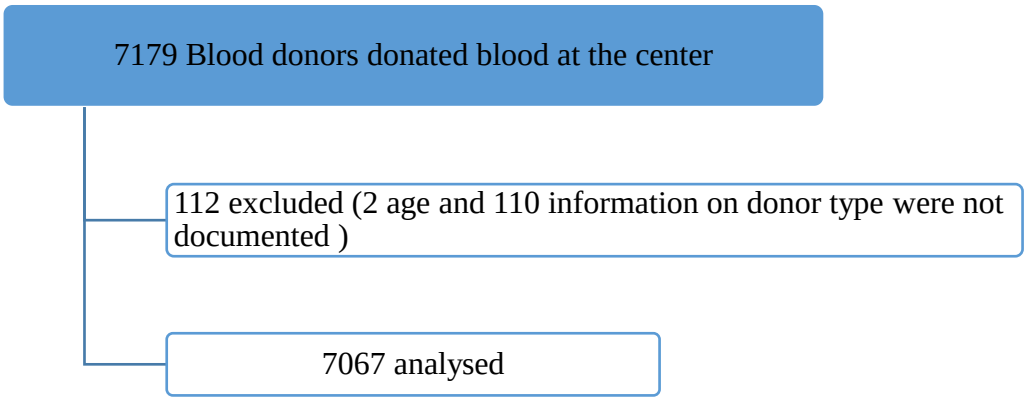


Figure 1: Study participant recruitment

The predominant age group was 25 to 49 years, and the mean age (±SD) was 37.08(±10.07). Males comprised 6423 (90.9%) of the study participants, and most blood donors were from Ilala municipality. Our findings showed that 4892 (69.2%) were voluntary blood donors, and first-time donors comprised 4631 (65.5%) of the study participants (Table 1).

Table 1: Socio-demographic characteristics of the study participants (N=7067)

Characteristics	Categories	Number (%)
Age(years)	18-24	702 (9.9)
	25-49	5413 (76.6)
	50-65	952 (13.5)
Sex	Male	6423 (90.9)
	Females	644 (9.1)
Residence	Ilala	3572 (50.5)
	Temeke	1214 (17.2)
	Kinondoni	1500 (21.2)
	Ubungo	456 (6.5)
	Kigamboni	84 (1.2)

	Out of Dar es Salaam	241(3.4)
Blood donor type	Family replacement donors	2175 (30.8)
	Voluntary donors	4892 (69.2)
Blood donation frequency	First-time donor	4631 (65.5)
	Repeat blood donor	2436 (34.5)

Prevalence of Hepatitis B infection among blood donors

Out of 7067 donors who donated blood, 383 (5.4%) tested positive for HBV infection (Figure 2).

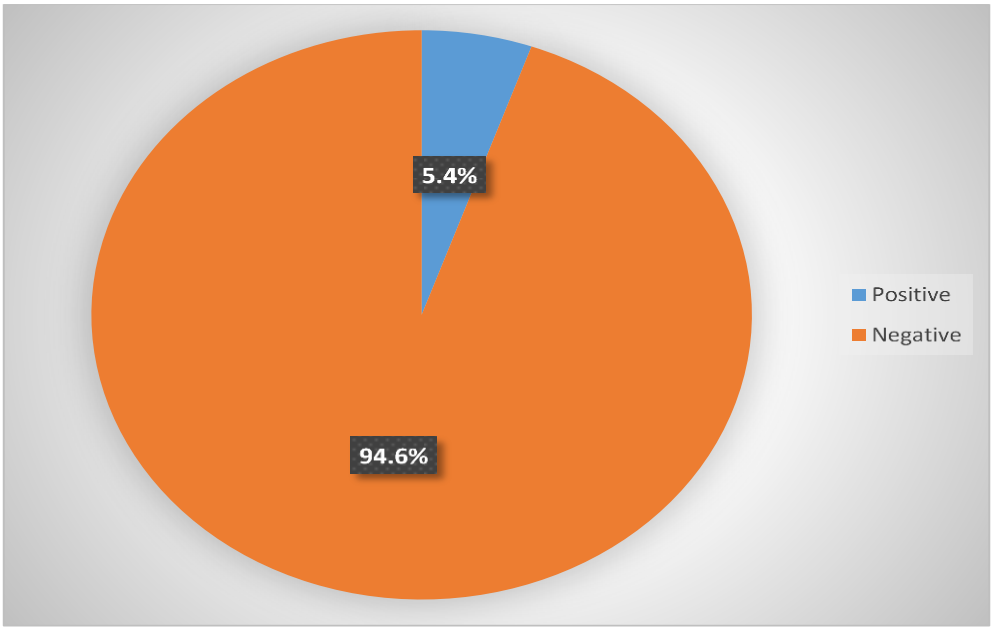


Figure 2: Prevalence HBV infection among blood donors

In univariate analysis, blood donor type and blood donation frequency were associated with HBV infection among blood donors (P < 0.001 and p = 0.006, respectively) (Table 2).

Table 2: Prevalence of HBV infection by sociodemographic characteristics of the study participants (N=7069)

Characteristics	Attribute	Total donors (N=7069)	HBV-infected donors 383 (5.4%)	P value
Age(years)	18-24	702	35 (5.0)	0.637
	25-49	5413	301 (5.6)	
	50-65	952	47 (4.9)	
Sex	Females	644	43 (6.7)	0.139
	Male	6423	340 (5.3)	

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Place of residence	Ilala	3572	198 (5.5)	0.513
	Temeke	1500	88 (5.9)	
	Kinondoni	1214	62 (5.1)	
	Ubungo	456	22 (4.8)	
	Kigamboni	84	1 (1.2)	
	Out of Dar es Salaam	241	12 (5.0)	
Blood donor type	FRBD	2225	183 (8.2)	0.0001
	VNRBD	4842	200 (4.1)	
Blood donation frequency	First-time donor	4631	276 (6.0)	0.006
	Repeat blood donor	2436	107 (4.4)	

Factors with a P value <0.05 in the univariate analysis were considered for the multivariate analysis. FRBD predicted the occurrence of HBV infection, while first-time donation did not maintain statistical significance in the multivariate logistic regression. It was revealed that HBV infection was more likely (2.1x) to occur in FRBD (aOR=2.1, 95%CI= 2.1 (1.64-2.58), P < 0.001) compared to voluntary blood donors (Table 3).

Table 3: Multivariate analysis of factors associated with HBV infection among asymptomatic blood donors (N=7069)

Characteristics	Attribute	cOR (95%CI)	P-value for cOR	aOR (95%CI)	P-value for aOR
Blood donor type	FRBD	1.2 (0.97-1.50)	0.08	2.1 (1.64-2.58)	0.0001
	VNRBD	1		1	
Blood donation frequency	First-time donor	1.4 (1.10-1.73)	0.006	1.0 (0.80-1.32)	0.809
	Repeat blood donor	1		1	

Discussion

Blood transfusion safety remains a significant challenge in transfusion medicine, particularly in low- and middle-income countries. Our study highlights the factors associated with asymptomatic HBV infection among blood donors in Dar es Salaam. We conducted this study at the walk-in center in Eastern Zone, where the majority of blood donors were between 25 and 49 years old, comprising 76.6% of the study participants. Wahome et al., Abdella et al., and Kabemba et al. reported slightly younger ages compared to our study, which were 18-25 years (82%), 18-24 years (60%), and 18-38 years (80%), respectively [Wahome et al.2021, Abdella et al.2020, Kabemba et al.2017]. The older ages observed in our study is explained by the differences in the study setting, where it is likely to find younger people in educational institutions, while this study was conducted at a walk-in site.

Majority of study participants were male, similar to previous reports from Wongjarupong et al. and Mohamed et al. where most blood donors were males (71.7% and 84% respectively) [Wongjarupong et al.2021, Mohamed et al.2019].

The lower blood donation rate among females in our study can be a result of several factors, including inherently lower hemoglobin levels among females compared to male donors. At NBTS, the cutoff point for blood donation by hemoglobin status is 12.5g/dl for both males and females, while

normal Hb for females is 12.5-16.5g/dl, and that of males is 13.5-18.5g/dl thus a cut off of 12.5g/dl is likely to exclude more female donors than male counterpart [NBTS.2015].

The lower hemoglobin level among females is also attributed to monthly menstrual flow [Ofojekwu et al.2013]. Another reason could be the differences in donation rates between males and females; while males can donate blood every three months, females may only donate every four months, contributing to higher donation frequencies among males. Furthermore, during pregnancy and breastfeeding, females are not allowed to donate blood due to the increased demands of the growing fetus and neonate, leading to lower donation rates among females [NBTS.2015].

The majority of participants (64%) were voluntary blood donors, similar to reports by Abdella et al. (93.8%) and Kabemba et al. (83%) [Abdella et al.2020, Kabemba et al.2017]. However, a study by Mohamed et al. (2019), conducted at Muhimbili National Hospital, reported that 88% of donors were from FRBD [Mohamed et al.2019]. The difference in donor types between Mohamed et al. and our study arises from variations in the study setting, where FRBD donors are more likely to contribute for their sick relatives in a hospital environment.

This study, conducted in the eastern part of Tanzania, reports that the majority of participants (65.5%) were first-time donors. Another survey from western Tanzania by Marira et al. reported that 94.2% were first-time donors [Marira et al.2024]. Our findings differ from those of Al-Mohani et al., Ntawuyamara et al., and Mohamud et al., who reported first-time donation rates of 31.15%, 24.4%, and 58%, respectively [Ntawuyamara et al.,2024, Al-Mohani et al.2024, Mohamud et al.,2024]. Reliance on school donations and lower voluntary donation rates could explain the frequency of first-time donations in Tanzanian studies.

We observed an overall prevalence of Hepatitis B infection of 5.4%, while a prevalence of 6.9% was reported in Western Tanzania [Marira et al. 2024]. The difference between these two studies could be due to the various sources of reviewed donor records in Western Tanzania, which were taken from donor samples in hospital settings. Blood donors in the hospital environment are more likely to be family replacement donors and to donate under pressure, so they may not disclose risk factor information during pre-donation counseling. Studies in Cameroon and Mogadishu reported even higher prevalences (12.1% and 9.7%, respectively), with data sourced from hospital-based blood banks. Contrary to our findings, lower prevalence rates have been reported in Burundi (1.92%) and Brazil (0.62%) [Ntawuyamara et al. 2024, Gurwale et al. 2022]. The difference in HBV prevalence could be attributed to factors such as the general population prevalence of HBV; for instance, the general prevalence of HBV in Burundi was 4.6% [Kamali et al. 2021].

The predictors of HBV infection in our study were first-time donors and FRBDs. Similar findings have been reported from Burundi, where the prevalence of HBV infection was 3% among first-time donors compared to 1.2% among repeat donors [Ntawuyamara et al. 2024]. In Cameroon, a higher prevalence was reported among FRDs (7.6%) compared to 6.1% in VNRBDs, a trend similar to our findings [Ngomtcho et al. 2024]. A survey from Mogadishu also reported that first-time donors had a higher HBV prevalence than repeat blood donors [Mohamud et al. 2024]. The higher prevalence of HBV among first time donor could be because all repeat donors before subsequent donations receive post-donation counselling, and they are only allowed to donate blood if they did not react to transfusion transmissible infection from their previous donation

The primary limitation of the study was the small sample size, as we had to exclude blood units collected from sites other than the fixed site due to incomplete records in the BECS. In addition, clinical information such as recent febrile illness and other risk factors such as lifestyle were not captured in the database. Finally, information on blood donor occupation was not provided

Conclusion

Family replacement donations predicted a higher rate of HBV infections in this group compared to voluntary blood donation; therefore, the NBTS should enhance the application of donor selection guidelines and promote voluntary blood donation.

Recommendation

Based on our findings, we recommend the following: First, NBTS should invest in recruiting VNRBD and converting FRBD to increase the pool of VNRBDs. Next, NBTS should implement strategies for blood donor retention to boost the number of repeat blood donors. This can be achieved by adhering to national and international standards and guidelines on blood donor recruitment and retention.

Conflict of interest disclosures

The authors declare no conflicts of interest.

Author contributions

All authors contributed to the writing of the final manuscript and approved the final version to be submitted.

Data sharing statement

We will provide primary data access to the Tanzania Journal of Health Research when requested during the review process and/or for editorial decisions.

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